

Passband Signal, Complex Envelope, and Baseband Equivalent

A Unified Mathematical Treatment

1 Introduction

In communication theory, the same information-bearing waveform can be represented in three mathematically equivalent forms:

- Real passband signal
- Analytic signal using Hilbert transform
- Complex baseband (low-pass) equivalent, also called the **complex envelope**

Understanding the precise relationship between these forms is fundamental to modulation, demodulation, SDR, and DSP implementations.

2 Real Passband Signal

A real passband signal is centered around a carrier frequency f_c . This is the physical waveform transmitted over the air:

$$x_{PB}(t)$$

Because it is real-valued, its spectrum satisfies

$$X_{PB}(-f) = X_{PB}^*(f)$$

which means it contains symmetric positive and negative frequency components.

3 Analytic Signal and Hilbert Transform

To obtain a complex representation, we form the **analytic signal** using the Hilbert transform.

$$x_a(t) = x_{PB}(t) + j \mathcal{H}\{x_{PB}(t)\}$$

where $\mathcal{H}\{\cdot\}$ denotes the Hilbert transform.

Key property: The analytic signal contains only positive frequency components.

4 Complex Envelope (Formal Definition)

The complex envelope is obtained by removing the carrier from the analytic signal:

$$x_{CE}(t) = x_a(t) e^{-j2\pi f_c t}$$

This signal is low-pass in nature and contains all information of the original passband signal.

5 Complex Baseband Equivalent

The complex baseband signal is expressed as

$$x_{BB}(t) = I(t) + jQ(t)$$

The original passband signal can be reconstructed as

$$x_{PB}(t) = \text{Re} \left\{ x_{BB}(t) e^{j2\pi f_c t} \right\}$$

Hence,

$$x_{CE}(t) \equiv x_{BB}(t)$$

6 I/Q Interpretation

Let

$$x_{CE}(t) = I(t) + jQ(t)$$

Substituting into the reconstruction formula:

$$x_{PB}(t) = I(t) \cos(2\pi f_c t) - Q(t) \sin(2\pi f_c t)$$

This is the origin of quadrature modulation used in all modern communication systems.

7 Frequency Domain Interpretation

A real passband signal has symmetric spectrum. If we directly shift it to baseband, the negative frequencies overlap with positive ones, causing distortion.

The Hilbert transform removes the negative frequencies first, enabling proper baseband conversion.

8 Complete Relationship Chain

$$\begin{aligned}x_a(t) &= x_{PB}(t) + j\mathcal{H}\{x_{PB}(t)\} \\x_{CE}(t) &= x_a(t)e^{-j2\pi f_c t} \\x_{BB}(t) &= I(t) + jQ(t) \\x_{PB}(t) &= \text{Re}\{x_{BB}(t)e^{j2\pi f_c t}\}\end{aligned}$$

9 Practical Importance

- All SDR receivers convert passband signals to baseband using this principle.
- I/Q modulation and demodulation arise directly from this formulation.
- DSP simulations avoid carrier frequency by working in baseband.
- QAM, QPSK, OFDM are naturally described using complex baseband signals.

10 Key Insight

The complex envelope is the formally defined complex baseband equivalent obtained from the analytic signal of the passband waveform.